

Jinnah Medical - OT & ICU Module

Enhancement Guide

Executive Summary

This document outlines the recent enhancements made to the **Operating Theater (OT)** and **Intensive Care Unit (ICU)** modules, including new features, workflow improvements, and user guidance for effective management.

1. OPERATING THEATER (OT) MODULE

1.1 New Features Implemented

WHO Surgical Safety Checklist - Timeout Enforcement

What is it?

A mandatory safety protocol based on WHO guidelines that ensures patient safety before surgery begins.

Key Components:

- **Timeout Banner:** Visual alert on scheduled surgeries requiring timeout completion
- **Timeout Modal:** Interactive checklist with mandatory confirmations:
 - Patient identity verification
 - Procedure confirmation
 - Site/side marking verification
 - Antibiotic prophylaxis timing check (60-minute window)
- **Antibiotic Alert:** Warning if antibiotics given outside the safe window

Files Modified:

- `src/pages/hospital/ot/ot_Schedule.tsx` - Added timeout enforcement UI
- `src/features/hospital/ot/ot.api.ts` - API endpoints for timeout updates

Surgical Site Infection (SSI) Tracking System

What is it?

Complete tracking system for post-operative infections per CDC NHSN guidelines.

Infection Types Tracked:

- Superficial Incisional SSI
- Deep Incisional SSI
- Organ/Space SSI

Features:

- Case registration with risk stratification
- Procedure details capture
- Infection event logging with date tracking
- Outcome recording (resolved/expired/ongoing)
- Statistics dashboard with visual indicators

Files Created:

- `src/pages/hospital/ot/ot_SSITracking.tsx` - SSI management UI
- `backend/src/modules/hospital/models/OTSSITracking.ts` - Database schema
- `backend/src/modules/hospital/controllers/ot.controller.ts` - CRUD operations
- `backend/src/modules/hospital/routes/index.ts` - API routes

Files Modified:

- `src/pages/hospital/hospital_OTDashboard.tsx` - Added SSI tracking quick link

1.2 OT Module Workflow

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[Step 1] SCHEDULE SURGERY



- Navigate to OT > Schedule
- Create booking with patient, procedure, team details
- System assigns OT room and time slot



[Step 2] PRE-OPERATIVE PREPARATION



- Sterilization logs checked (OT > Sterilization)
- Equipment verified (OT > Equipment)
- Team assigned (OT > Team)



[Step 3] WHO TIMEOUT (MANDATORY)



- Click "TIMEOUT REQUIRED" banner on in-progress surgery
- Complete all checklist confirmations:

✓ Patient identity confirmed

✓ Procedure confirmed with consent

✓ Site/side marked and confirmed

✓ Antibiotics given within 60 min (if applicable)

- Click "Complete Timeout" to proceed



[Step 4] SURGERY EXECUTION



- Perform procedure
- Document in IPD Surgery Record



[Step 5] POST-OPERATIVE MONITORING



- Patient transferred to ICU/Recovery/Ward
- SSI tracking initiated if high-risk



[Step 6] SSI TRACKING (For 30 Days Post-Op)



- Monitor for infection signs
- Log any SSI events in OT > SSI Tracking
- Record outcomes and resolution

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1.3 User Tutorial - OT Management

How to Schedule a Surgery

1. **Navigate** to Hospital → OT → Schedule
2. **Click** "New Booking" button
3. **Enter** patient details (MRN or search by name)
4. **Select:**
 - OT Room
 - Surgery Date & Time
 - Procedure Type
 - Surgeon & Team
 - Anesthesia Type

1. **Click** "Save Booking"

How to Complete WHO Timeout

1. **Look for** "TIMEOUT REQUIRED" banner on in-progress surgeries
2. **Click** "Complete Timeout" button
3. **Verify** each checklist item:
 - Check "Patient identity confirmed"
 - Check "Procedure confirmed"
 - Check "Site/side marked"
 - Check "Antibiotics given" (if applicable)
1. **Note:** If antibiotics were given >60 minutes ago, a warning will appear
2. **Click** "Complete Timeout" to finalize

How to Track SSI (Surgical Site Infections)

1. **Navigate** to Hospital → OT → SSI Tracking (or click from OT Dashboard)
2. **View** active cases in "Active Monitoring" tab
3. **To Add New Case:**

- Click "Add Case"
- Select patient from surgery history
- Enter procedure details and risk factors
- Set monitoring start date

1. **To Record Infection Event:**

- Find patient in case list
- Click "Record Event"
- Select infection type (Superficial/Deep/Organ)
- Enter detection date and symptoms

1. **To Update Outcome:**

- Click on case
- Update status: Resolved / Expired / Ongoing
- Enter resolution date if applicable

How to Manage OT Dashboard

1. **View Room Status:** See which OT rooms are occupied/available
2. **Quick Actions:**
 - Click "Schedule" to create bookings
 - Click "SSI Tracking" for infection monitoring
 - Click "Sterilization" for equipment logs
 - Click "Team" for staff assignments
1. **Monitor:** Check timeout status indicators on active surgeries

2. INTENSIVE CARE UNIT (ICU) MODULE

2.1 Bug Fixes & Improvements

Issues Resolved:

- Fixed API call for ward listing (listIPDWards → listWards)

- Fixed type error in discharge status handling
- Removed invalid parameter from ventilator patient listing

Files Modified:

- src/pages/hospital/icu/icu_Transfer.tsx - Fixed ward loading and discharge types
- src/pages/hospital/icu/icu_Ventilator.tsx - Fixed admission filtering

2.2 ICU Module Workflow

ICU MODULE WORKFLOW |

[Step 1] PATIENT ADMISSION

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- Patient transferred from OT/ER/OPD

- Bed assignment in ICU
- Initial vitals and assessment recorded

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[Step 2] MONITORING & CARE

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- Continuous vitals monitoring
- Ventilator management (if required)
- Medication administration
- Daily progress notes

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[Step 3] TRANSFER/DISCHARGE DECISION

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- Patient stabilized → Transfer to ward
- Requires ventilation → Continue ICU care
- Condition worsened → Re-evaluate treatment
- Expired → Record death certificate

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[Step 4] TRANSFER EXECUTION

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- Navigate to ICU > Transfer
- Select patient
- Choose destination:
 - General Ward
 - Home (Discharge)
 - Other Hospital
 - Death (Deceased)

- Enter summary and date/time

- Confirm transfer

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[Step 5] VENTILATOR MANAGEMENT

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- For ventilated patients:
 - Monitor ventilator settings
 - Record weaning attempts
 - Track duration and outcomes

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2.3 User Tutorial - ICU Management

How to Admit a Patient to ICU

1. **Navigate** to Hospital → ICU → Admissions
2. **Click** "New Admission"
3. **Search** patient by MRN or name
4. **Select** available ICU bed
5. **Enter**:
 - Admission date/time
 - Admitting doctor
 - Reason for admission
 - Initial vitals
 - Ventilator required? (Yes/No)
1. **Click** "Admit Patient"

How to Transfer/Discharge from ICU

1. **Navigate** to Hospital → ICU → Transfer
2. **Select** patient from admission list
3. **Choose** destination:
 - **Ward**: Transfer to general ward (select specific ward)
 - **Home**: Patient discharged
 - **Other Hospital**: Transfer to external facility
 - **Deceased**: Record patient death
1. **Enter**:
 - Transfer/discharge date & time
 - Summary of ICU stay
 - Destination details

1. **Click** "Confirm Transfer"

How to Manage Ventilator Patients

1. **Navigate** to Hospital → ICU → Ventilator
2. **View** list of patients currently on ventilator
3. **Select** patient to view details:

- Ventilator settings
- Duration on ventilator
- Weaning attempts

1. **Update Settings** (if authorized):

- Mode (SIMV, CMV, etc.)
- FiO2, PEEP, Rate
- Tidal volume

1. **Record Weaning Attempt:**

- Date/time of attempt
- Settings used
- Patient tolerance
- Outcome (Successful/Failed)

How to View ICU Dashboard

1. **Access** Hospital → ICU → Dashboard
2. **Monitor:**

- Total occupied beds
- Available beds
- Ventilator utilization
- Patient acuity levels

1. **Quick Actions:**

- Click bed number for patient details
 - Color indicators show severity (Green/Yellow/Red)
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3. QUICK REFERENCE - COMMON TASKS

Emergency Surgery Override

When to use: Life-threatening emergency requiring immediate surgery without full timeout

Steps:

1. Create emergency booking in OT Schedule
2. Mark as "Emergency" priority
3. Timeout modal will show "Emergency Override" option
4. Document reason for override
5. Complete minimal safety checks possible
6. Proceed with surgery

SSI Risk Stratification

High-Risk Indicators:

- Dirty/Infected wound class
- ASA score > 3
- Emergency procedure
- Diabetes/Immunocompromised patient
- Long operation duration (>2 hours)

Action: These cases are auto-flagged for enhanced SSI monitoring

ICU Bed Management

Bed Status Colors:

-  Green: Available
-  Yellow: Occupied - Stable
-  Red: Occupied - Critical
-  Gray: Under maintenance

Transfer Protocol:

1. Always check bed availability at destination
 2. Ensure handover notes are complete
 3. Transfer with nurse escort for critical patients
 4. Update system immediately after transfer
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4. BEST PRACTICES

OT Module

DO:

- Complete timeout before first incision
- Document antibiotic timing accurately
- Track all high-risk patients for SSI
- Update SSI status within 24 hours of detection

DON'T:

- Skip timeout for non-emergency cases
- Delay SSI documentation beyond 48 hours
- Forget to mark site/side before timeout

ICU Module

DO:

- Record vitals every hour for critical patients
- Update ventilator settings immediately after changes

- Document weaning attempts with detailed notes
- Transfer patients only after stabilization

✗ DON'T:

- Leave ventilator alarms unattended
- Delay discharge documentation
- Forget to record reason for ICU admission

5. SUPPORT & TROUBLESHOOTING

Common Issues

Issue	Solution
Timeout won't complete	Check all 3 confirmation boxes are checked
Can't find patient for SSI tracking	Ensure patient has completed surgery record
ICU bed not showing available	Check for pending transfers not completed
Ventilator patient not in list	Verify "ventilator required" was checked at admission

Contact Information

For technical issues or feature requests:

- **IT Support:** ext. 5000
- **System Admin:** ext. 5001

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